LESSON PLAN

LSF, STABILITY, A/C PERFORMANCE (6)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	15
Total No. of Lesson Plans	33	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Lift augmentation
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Lift augmentation. High lift devices. Boundary Layer control
5. Specific Objectives	6.1.2.1 Trainee will be able to describe various types of aerofoil sections, conditions of performance, theories, and concept related with performance

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of IB diagrams regarding flap deflection angles and boundary layer separation points; Instructor supervision of diagram interpretation.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of high lift devices?
(b) How does increasing the angle of attack affect the lift coefficient?
(c) Define the boundary layer in the context of wing performance.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Types of High Lift Devices • Slats and Leading Edge Flaps • Trailing Edge Flaps • Flap Deflection Angles	L & Ds	PPT, IB	15 min
(b)	• Boundary Layer Control Methods • Vortex Generators Function • Blowing and Suction Systems • Boundary Layer Transition	L & Ds	PPT, PDF	20 min
(c)	• Performance Impact of Augmentation • Stall Speed Reduction • Takeoff and Landing Efficiency • Drag Penalty Considerations	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how a leading edge slat delays flow separation.
(b) Describe the trade-off between lift augmentation and drag increase.
(c) Identify two methods used for active boundary layer control.

12. Home Assignment and Follow Up	Prepare a brief note comparing the operational advantages of slats versus trailing edge flaps for takeoff and landing phases, referencing LP 16 concepts on control surface limitations.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Secondary/Auxiliary flight controls, thrust reversing/braking	Refer to the master syllabus reference list	LP 16

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min